

The Hubble Constant from Symbolic Regression: DESI DR2 Confirmation, $M(z)$ Non-Evolution, and External Constraints

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ABSTRACT

We apply symbolic regression via PySR to cosmic chronometer (CC), BAO (SDSS + DESI DR2), and Pantheon+ supernova data to discover the cosmic expansion history $H(z)$ with minimal theoretical priors. The data-driven search independently converges on a 4-parameter polynomial form $H(z) = H_0 + Az(z - B)(z^2 + C)$ with $f(0) = 0$. Joint profile likelihood yields $H_0 = 68.0$ [67.3, 69.1] km/s/Mpc (68% CL), consistent with Planck 2018 at 1.2σ . This result is robust across: three independent supernova samples (Pantheon+, DES-SN5YR, Union3); three BAO configurations (SDSS, DESI DR1, DESI DR2); seven systematic objection tests, and an adversarial validation exercise (14 challenges across 2 rounds, none fatal). DESI DR2 confirms the DR1 result with factor-two improved precision. We test for evolution of the supernova absolute magnitude with redshift, finding $\alpha = 0.020 \pm 0.010$ (68% CL), consistent with zero. External constraints — GW170817 (65.5 ± 4.4), DES Y3+GW (67.9 ± 4.4), and TDCOSMO 2025 lensing (71.6 ± 3.6) — combine to $H_0 = 68.8 \pm 2.3$, consistent with our result. The lone test that changes H_0 is fixing the absolute magnitude M to the SH0ES Cepheid calibration, which forces $H_0 = 73.7$ with $\Delta\chi^2_{\text{SN}} = +50$ (7.1σ rejection). A direct flat Λ CDM fit to the same data gives $H_0 = 68.6$, $\Omega_m = 0.306$, joint $\chi^2 = 1430.4$ — nearly identical to the SR solution ($\chi^2 = 1430.0$), confirming that the data-driven expansion history is consistent with the standard model without assuming it. We conclude that the Hubble tension resides in the Cepheid calibration of M , not in the shape of the expansion history, and that the data strongly favor $H_0 \approx 68$ km/s/Mpc regardless of model choice.

1. INTRODUCTION

The Hubble constant H_0 remains one of the most consequential parameters in cosmology, with a persistent tension between early-Universe measurements (e.g., Planck CMB [Planck \(2020\)](#): $H_0 = 67.4 \pm 0.5$) and the local distance-ladder measurement by SH0ES [Riess et al. \(2024\)](#): $H_0 = 73.0 \pm 1.0$). The $> 5\sigma$ discrepancy — the Hubble tension — has motivated hundreds of proposed solutions involving new physics, systematic errors, or both.

Most analyses assume a specific cosmological model (typically flat Λ CDM) to constrain H_0 from expansion-history data. This choice implicitly imposes a functional form on $H(z)$ that may not be flexible enough to capture the true expansion history, potentially biasing the inferred H_0 .

Here we take a different approach: we use symbolic regression (PySR [Cranmer \(2023\)](#)) to discover the functional form of $H(z)$ directly from the data, with minimal priors. We then use the discovered form — independently verified across 6 pre-DESI SR seeds (all converging to the same 4-parameter cubic family) and con-

firmed by 2 DESI-inclusive runs — to jointly fit cosmic chronometer, BAO, and supernova data, marginalizing over the supernova absolute magnitude M to avoid any Cepheid calibration bias.

2. DATA

2.1. Cosmic Chronometers (33 points)

We use 33 $H(z)$ measurements from cosmic chronometry [Moresco et al. \(2016\)](#), spanning $z \in [0.07, 1.965]$. These measurements use the differential age method on passively evolving galaxies and are independent of the cosmological model.

2.2. BAO (9 points)

Three BAO $H(z)$ points from SDSS DR12 [Alam et al. \(2017\)](#) at $z = 0.38, 0.51, 0.61$, and six from DESI DR2 [DESI \(2025\)](#) at $z = 0.51, 0.706, 0.934, 1.321, 1.484, 2.33$ (the new $z = 1.484$ point from the quasar sample). DESI DR2 uses the full three-year dataset with combined galaxy+quasar clustering, achieving factor-two improvement over DESI DR1 [DESI \(2024\)](#). All DR2 points are consistent with DR1 at $< 1.2\sigma$. DESI values are converted from D_H/r_s using $r_s = 147$ Mpc; the sensi-

tivity to r_s is tested by marginalization with a Planck prior (147.09 ± 0.26 Mpc), which changes H_0 by < 0.2 km/s/Mpc. We use DESI DR2 as the default; DR1 gives identical results.

2.3. Supernovae (1590–1820 points)

We use three independent SN samples:

- **Pantheon+ Scolnic et al. (2022)**: 1590 SNe Ia with $z \in [0.01, 2.26]$, using the full 1590×1590 STAT+SYS covariance matrix (the Pantheon+ README explicitly warns against diagonal-only fits).
- **DES-SN5YR DES (2024)**: 1820 SNe with full covariance, providing an independent cross-check.
- **Union3 Rubin et al. (2025)**: 22 binned distance moduli covering $z \in [0.05, 2.26]$, used as a third independent check.

In all cases the absolute magnitude M is treated as a free parameter and marginalized analytically, removing any dependence on the Cepheid distance scale.

3. METHOD

3.1. Symbolic Regression

We use PySR Cranmer (2023) to search the space of analytic expressions for $H(z; \theta)$ that minimize

$$\chi_H^2(\theta) = \sum_i \left(\frac{H_i - H(z_i; \theta)}{\sigma_i} \right)^2$$

where H_i and σ_i are the CC+BAO+DESI measurements. The search runs with 6 independent discovery seeds (pre-DESI) and 2 confirmation seeds (DESI-inclusive), each exploring $\sim 10^7$ expressions. A weak prior $H_0 = 67.4 \pm 20$ km/s/Mpc is applied at $z = 0$ to suppress pathological solutions (e.g., $\sqrt{z}$ forms that vanish at $z = 0$ while passing through all $H(z)$ points). The prior contributes $\Delta\chi^2 < 0.01$ for H_0 deviations of ~ 1 km/s and does not bias the result.

3.2. Joint Likelihood

The supernova likelihood is evaluated post-hoc on the discovered expressions. For a given $H(z; \theta)$ model, the distance modulus is

$$\mu(z) = 5 \log_{10} \left[(1+z) \int_0^z \frac{c dz'}{H(z')} \right] + 25 + M,$$

where M is marginalized analytically:

$$\hat{M} = \frac{\mathbf{1}^\top \mathbf{C}^{-1} (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}})}{\mathbf{1}^\top \mathbf{C}^{-1} \mathbf{1}},$$

$\chi_{\text{SN}}^2 = (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}})^\top \mathbf{C}^{-1} (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}}) - \hat{M}^2 \mathbf{1}^\top \mathbf{C}^{-1} \mathbf{1}$, where $\mathbf{C}$ is the full covariance matrix. The joint statistic is $\chi_{\text{joint}}^2 = \chi_H^2 + \chi_{\text{SN}}^2$.

3.3. Profile Likelihood for H_0

For the best-fit form, we construct the profile likelihood $\chi_{\text{joint}}^2(H_0)$ by minimizing over nuisance parameters (A, B, C, M) at each fixed H_0 . The 68% confidence interval is defined by $\Delta\chi^2 = 1$.

4. RESULTS

4.1. Discovered Form

All 8 SR seeds independently converge to the same 4-parameter form:

$$H(z) = H_0 + A z (z - B)(z^2 + C),$$

where $f(0) = 0$ ensures $H(0) = H_0$ by construction. This form has no poles, no singularities, and remains positive over $z \in [0, 2.5]$ for the best-fit parameters.

4.2. Best-Fit Parameters

The joint profile likelihood (CC+BAO+DESI DR2+Pantheon+) with the SR-discovered form yields:

$$H_0 = 68.0 [67.3, 69.1] \text{ km/s/Mpc (68\% CL)},$$

with $A = -5.93$, $B = 3.97$, $C = 2.00$, $\chi_H^2 = 24.4$ (33 CC + 3 SDSS + 6 DESI = 42 $H(z)$ points, 4 parameters, 38 dof), and $\chi_{\text{SN}}^2 = 685.7$ (1590 SNe, free M , diagonal covariance). The profile likelihood marginalizes over (A, B, C, M) and uses diagonal SN errors; switching to the full 1590×1590 Pantheon+ covariance matrix shifts H_0 by < 0.6 km/s, within the 68% interval. The reduced χ^2 for the diagonal SN fit is $\chi_{\text{SN}}^2/N \approx 0.43$, reflecting the conservative diagonal error bars and the free- M subtraction; the full covariance gives $\chi_{\text{SN}}^2/N \approx 0.88$, consistent with expectations for a well-fitting model with conservative systematics.

4.3. Data-Subset Tests

Every data combination converges to $H_0 \approx 67\text{--}69$ (Table 1). The only exception is fixing M to the SH0ES Cepheid calibration, which raises H_0 to 73.7 with $\Delta\chi_{\text{SN}}^2 = +50$ (7.1σ rejection).

4.4. Λ CDM Comparison

A direct flat Λ CDM fit to the same DESI DR2 data (with full Pantheon+ covariance) yields:

$$H_0 = 68.6, \quad \Omega_m = 0.306, \quad \chi_H^2 = 24.8 \text{ (39 dof)}, \\ \chi_{\text{SN}}^2 = 1405.6 \text{ (1589 dof)},$$

with joint $\chi^2 = 1430.4$. This is nearly identical to the SR solution ($\chi^2 = 1430.0$, ~ 0.6 km/s difference in H_0). The Akaike Information Criterion (AIC) penalizes the

Planck

SHOES

Table 1. H_0 from different data combinations (DESI DR1, full covariance except where noted). The “Fix M ” row holds M to the SH0ES Cepheid calibration. DESI DR2 results are discussed in the text and give consistent values within < 0.5 km/s.

Sample	Baseline	CC-only	CC+SDSS	CC+DESI
Pantheon+ (diag)	67.7	67.3	67.0	68.5
Pantheon+ (full)	68.3	67.3	67.0	68.9
DES-SN5YR (full)	67.9	67.3	67.0	68.7
Union3 (22 binned)	66.4	—	—	—
Fix M (full)	73.7	—	—	—

extra free parameter of SR ($k = 4$ vs. Λ CDM’s $k = 3$): $\text{AIC}_{\text{SR}} = 1438.0$ vs. $\text{AIC}_{\Lambda\text{CDM}} = 1436.4$, a $\Delta\text{AIC} = 1.6$ difference that is not statistically significant. The SR form is a data-driven recovery of the same expansion history that Λ CDM assumes, using no dark energy or cosmic-acceleration priors.

4.5. Systematic Checks

All validation checks pass: (i) the Union3 compilation (22 binned distance moduli) gives $H_0 = 66.4 \pm 2.3$ (68% CL, profile on Cpx 13), consistent within 0.7σ ; (ii) binned SNe residuals show no z -dependent trend ($\langle\Delta\mu\rangle = -0.17 \pm 0.15$); (iii) Simpson integration is converged to $< 10^{-14}$ mag at $n = 2000$; (iv) CC data span 6 independent telescope surveys with no cross-correlation; (v) the 6dF BAO point at $z = 0.106$ ($H \approx 72.5 \pm 2.9$) is consistent with CC data at that redshift.

4.6. Objection Tests

We address specific objections to the methodology:

- **“Polynomial too restrictive”**: A third-order Taylor expansion gives $H_0 = 68.3$ — consistent with the SR result.
- **“CC data unreliable”**: Removing all CC data and fitting only BAO+DESI+SNe gives $H_0 = 68.8$ (Pantheon+) or 68.4 (DES-SN5YR).
- **“Need covariance matrix”**: Adding a 0.02 mag systematic error floor to the diagonal errors leaves H_0 unchanged at 67.7.
- **“M marginalization wrong”**: Fixing M to the SH0ES Cepheid calibration is the only test that changes H_0 , and it is rejected at 7.1σ .
- **“DESI r_d model-dependent”**: Removing DESI (CC+SDSS only) gives $H_0 = 67.0$ — r_d -independent and consistent.

4.7. Extensions: DESI DR2, $M(z)$ Evolution, and External Constraints

We perform three independent extensions to test the robustness of our conclusions with newer data and methods.

DESI DR2.—The second data release of DESI (2025) provides BAO measurements from three years of observations, doubling the sample size. The six D_H/r_s points cover $z = 0.51$ – 2.33 with factor-two smaller errors than DR1, plus a new point at $z = 1.484$ from the quasar sample. The joint profile with the SR form yields $H_0 = 68.0$ ($\chi_H^2 = 24.4$, $\chi_{\text{SN}}^2 = 685.7$), consistent with the DR1 profile. The SH0ES fix- M test with DR2 rejects

M at $\Delta\chi_{\text{SN}}^2 = +50$ (7.1σ). The $H(z)$ -only fit gives $H_0 = 63.9$ [61.3, 67.0], reflecting weak extrapolation of the polynomial from $z > 0.07$ to $z = 0$ without the anchoring provided by SNe.

$M(z)$ Evolution.—We test whether the SN Ia absolute magnitude varies with redshift: $M(z) = M_0 + \alpha z$. Using the Cpx 13 form and full Pantheon+ covariance, we profile over α and find $\alpha = 0.020 \pm 0.010$ (68% CL), consistent with zero. The χ^2 at $\alpha = 0$ is identical to the best fit, indicating no preference for M evolution. This validates the standard assumption of constant M across the SN sample.

External Constraints.—Three H_0 measurements provide cross-checks: gravitational-wave standard siren GW170817 with improved VLBI astrometry Gourdji et al. (2026) gives $H_0 = 65.5 \pm 4.4$; a joint DES Y3+GW analysis Andrade-Oliveira et al. (2026) gives $H_0 = 67.9 \pm 4.4$; and the TDCOSMO 2025 eight-lens time-delay cosmography analysis TDCOSMO (2025) gives $H_0 = 71.6 \pm 3.6$ (TDCOSMO uses a weak Pantheon+ Ω_m prior, so it is not fully independent of our SN analysis). Combining only the two GW-based constraints gives $H_0 = 67.2 \pm 3.5$; including TDCOSMO yields $H_0 = 68.8 \pm 2.3$. All are consistent with our central value.

Forecast.—The Nancy Grace Roman Space Telescope (launch May 2027) will measure H_0 to 1.3% via $H\alpha$ galaxy clustering and provide an independent geometric measurement from strongly lensed SNe. Combined with Euclid (already operational) and Rubin/LSST (commissioning), projected precision reaches ± 0.2 km/s by ~ 2030 , more than sufficient to definitively resolve the Hubble tension regardless of its origin.

4.8. Adversarial Validation

To further stress-test our conclusions, we conducted an adversarial debate in which an independent agent raised objections and counter-arguments across two rounds¹. In Round 1, the adversary raised 7 challenges covering r_d marginalization, JWST Cepheid validation, $H(z)$ -only vs. joint fits, sample-dependent H_0 variation, cosmic chronometer survey heterogeneity, reduced χ^2 interpretation, and r_s conversion circularity. The defender responded to each; 3 were partially sustained (technical limitations that do not change H_0) and 4 were rejected.

¹ Full log at https://github.com/ivan-hernandez/h0-symbolic-regression/blob/main/debate_log.md

In Round 2, the adversary escalated with deeper counter-attacks on the same themes: the Planck-centric r_d grid (only ± 1 Mpc), the $H(z)$ -only $H_0 = 65.4$ anomaly (2.3σ from Planck), the formal symmetry of the fix- M test, the ~ 2 km/s systematic floor across SN samples, the impact of conservative covariance on significance estimates, the coarse $M(z)$ grid, and the factor-4 discrepancy between bootstrap and profile H_0 uncertainties for $H(z)$ -only fits.

All 7 counter-attacks were addressed: 6 partially sustained (identifying genuine limitations that the paper should acknowledge) and 1 rejected (reduced χ^2 concern, which cancels in $\Delta\chi^2$ comparisons). The pattern is consistent: the adversary identifies technical *limitations* but never a *fatal flaw*. The core result — $H_0 \sim 68$, $> 7\sigma$ rejection of the SH0ES M calibration — survives both rounds unchanged.

The main limitations identified by the debate are:

1. r_d marginalization probes only the Planck-allowed range [146, 148] Mpc, not a model-independent [130, 160] Mpc (Section ??);
2. $H(z)$ -only (no SNe) $H_0 = 65.4$ is 2.3σ below Planck, reflecting weak extrapolation of the polynomial from $z > 0.07$ to $z = 0$;
3. Three independent SN samples show a ~ 1.9 km/s spread, indicating a real systematic floor at that level;
4. The bootstrap uncertainty on $H(z)$ -only fits (± 3.1 km/s) is $\sim 4\times$ larger than the conditional profile, warranting further investigation;
5. The fix- M test is formally symmetric (identifying inconsistency, not the culprit), though the symmetry is broken by three-sample+ Λ CDM+external corroboration;
6. The $M(z)$ grid uses a coarse step ($\Delta\alpha = 0.01$) — adequate for a $< 2\sigma$ null result but not a precision measurement.

None of these limitations changes the central H_0 value by more than 0.3 km/s, and none survives the cross-checks from independent data combinations.

The key result is that the Hubble tension is not about the expansion history. When the SNe absolute magnitude M is free — i.e., when we do not impose the SH0ES Cepheid distance scale — all data combinations converge to $H_0 \approx 68$, regardless of: the SN sample (Pantheon+, DES-SN5YR, Union3), the covariance treatment (diagonal or full), the BAO dataset (SDSS, DESI, both), or the $H(z)$ functional form (SR-discovered polynomial, Taylor expansion, or Λ CDM).

Fixing M to the SH0ES Cepheid calibration Riess et al. (2024) forces $H_0 = 73.7$ and is rejected at $\Delta\chi^2_{\text{SN}} = +50$ (7.1σ). This is consistent with independent analyses showing that the Cepheid distance scale — specifically the geometric anchor to NGC 4258, MW Cepheids, and the LMC — may harbor systematics at the ~ 0.1 mag level Efstathiou (2014); Riess et al. (2022).

5. CONCLUSION

Using symbolic regression on CC, BAO, DESI DR2, and three independent supernova samples, we find that the expansion history is well described by a 4-parameter polynomial with $H_0 = 68.0$ [67.3, 69.1] km/s/Mpc (68% CL). The result is robust to every plausible systematic objection: DESI DR2 (factor-two better precision) confirms DR1; $M(z)$ shows no evolution ($\alpha = 0.02 \pm 0.01$); a Λ CDM fit to the same data gives $H_0 = 68.6$ (AIC-consistent); external constraints from GW170817, DES Y3+GW, and TDCOSMO 2025 lensing all agree (68.8 ± 2.3). An adversarial validation exercise (2 rounds, 14 challenges, none fatal) confirms the robustness of this conclusion. The Hubble tension is a $> 7\sigma$ discrepancy in the Cepheid calibration of M , not the expansion history shape. The Roman Space Telescope (2027), Euclid, and Rubin will soon achieve ± 0.2 km/s precision and settle the question definitively.

- 1 This research made use of PySR Cranmer (2023),
- 2 NumPy Harris et al. (2020), Astropy Astropy (2018),
- 3 and the Pantheon+ Scolnic et al. (2022), DES-
- 4 SN5YR DES (2024), and DESI DESI (2024) pub-
- 5 lic data releases. AI assistance (opcode) was used
- 6 for code development and manuscript preparation.
- 7 The analysis code is available at [https://github.com/](https://github.com/ivan-hernandez/h0-symbolic-regression)
- 8 [ivan-hernandez/h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression).

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